

REVIEW

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Ethical integration of generative artificial intelligence in Islamic religious education involves human mediation and source verification

Shahazwan Mat Yusoff^{1*} , Anwar Farhan Mohamad Marzaini¹, Siti Nadya Zynuddin², Ahmad Muhaimin Mat Jusoh², Mohd Helme Basal³, Nur Riza Alias⁴ and Norhashimi Saad¹

*Correspondence:

Shahazwan Mat Yusoff
shahazwan@um.edu.my

¹Department of Curriculum & Instructional Technology, Faculty of Education, Universiti Malaya, Kuala Lumpur, Malaysia

²Department of Educational Management, Planning and Policy, Faculty of Education, Universiti Malaya, Kuala Lumpur, Malaysia

³Faculty of Sports Science and Recreation, Universiti Teknologi MARA, Shah Alam, Malaysia

⁴Department of Language and Literacy Education, Faculty of Education, Universiti Malaya, Kuala Lumpur, Malaysia

Abstract

The present systematic literature review investigates the ethical integration of Generative Artificial Intelligence (GenAI) in Islamic Religious Education (IRE). In this review, ethical integration extends beyond general AI ethics principles such as privacy, transparency, academic integrity, and accountability by requiring GenAI use to remain consistent with Islamic educational values, source authenticity, teacher mediation, and religious and epistemic authority. Using PRISMA 2020 guidelines, this systematic literature review searched peer-reviewed English-language journal articles published between 2022 and 2026 in Scopus and Web of Science. Of 124 records initially identified, 13 studies were included in the synthesis. Thematic analysis using deductive and inductive coding revealed four main themes. First, GenAI in IRE is primarily discussed as a pedagogical assistant for lesson planning, teaching-material development, assessment support, personalised learning, administrative activities, and Qur'anic learning rather than as a replacement for teachers or religious authorities. Second, the literature highlights ethical, epistemic, and value-alignment concerns, including plagiarism, hallucinations, privacy issues, bias, unreliable sources, overdependence, doctrinal inaccuracies, and confusion regarding religious authority. Third, responsible integration requires multilayered safeguards, including teacher supervision, verification of AI-generated information, institutional policies, AI literacy, academic-integrity guidance, culturally sensitive use, and alignment with Islamic educational values. Fourth, the evidence base remains limited, fragmented, and predominantly conceptual, exploratory, or perception-based, with few classroom-based, longitudinal, comparative, or intervention studies. Accordingly, the findings should be interpreted as an emerging synthesis rather than conclusive evidence of the effects of GenAI in IRE. The review further identifies *adab*, *amanah*, and epistemic authority as complementary conceptual lenses because they respectively address appropriate conduct towards knowledge, responsible and trustworthy use of AI and knowledge, and the legitimate validation and interpretation of religious knowledge. Therefore, GenAI may support IRE when used as an assistive technology under



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appropriate human supervision and verification. Future research should develop and empirically examine Islamic ethical frameworks grounded in these dimensions.

Keywords Generative artificial intelligence, Islamic religious education, Ethical integration, Educational technology

1 Introduction

The rapid development of Generative Artificial Intelligence (GenAI) tools has altered contemporary discourse around education, particularly in content creation, personalised learning, feedback, assessment, instructional design, and administrative assistance. In educational settings, technologies such as ChatGPT cannot be considered solely in terms of their technological functions; they must also be understood in relation to the production, interpretation, access to, and assessment of knowledge. Previous research on artificial intelligence in education has demonstrated that AI-assisted technologies can support adaptive learning, assessment, intelligent tutoring, and academic services while also raising issues related to pedagogy, ethics, regulation, and the role of educators [45]. Recent literature on GenAI and large language models highlights their potential to assist teachers and students while emphasising critical thinking, fact-checking, human monitoring, and institution-specific strategies for responsible implementation [24, 27]. At the same time, the development of GenAI technologies has progressed more rapidly than the development of ethical guidelines and regulatory frameworks capable of ensuring responsible implementation across educational contexts. Consequently, contemporary scholarship identifies both the educational potential of GenAI and concerns related to misinformation, academic integrity, misuse, and governance [20, 30, 34]. These concerns become particularly significant when GenAI is introduced into educational domains in which knowledge is closely connected to religious belief, moral formation, and recognised sources of authority.

In this paper, Islamic Religious Education (IRE) refers to structured educational processes concerned with Islamic knowledge, Qur'anic learning, religious beliefs, worship, ethics, moral formation, and the education of Muslim learners. It includes formal and semi-formal processes in which Islamic teachings are mediated through teachers, institutions, curricula, and recognised religious sources. Islamic education has been conceptualised as a holistic educational process encompassing intellectual, moral, spiritual, and social dimensions of human development [19, 39]. IRE therefore differs not only from secular educational domains but also from religious education traditions grounded in other theological and epistemological traditions. While Christian, Jewish, and other forms of religious education may similarly involve moral formation, sacred texts, religious identity, and forms of religious authority, IRE is situated within an Islamic conception of knowledge and education in which the relationship between knowledge, ethical conduct, religious sources, teacher mediation, and moral formation is shaped by the Islamic worldview. In this sense, the distinctive concern of IRE is not simply that AI-generated information should be accurate, but that religious knowledge should be approached, transmitted, interpreted, and evaluated in ways that remain consistent with Islamic educational purposes and recognised sources of knowledge. This distinction is important because the present review draws selectively on broader religious education literature while treating such evidence as contextual rather than automatically transferable to IRE.

Recent studies indicate that AI and ChatGPT have been explored in Islamic education for lesson preparation, student-centred teaching, innovative pedagogy, self-inquiry, content creation, and educator assistance [4, 28, 40]. In IRE, reported or proposed applications include the preparation of teaching materials, development of learning activities and assessments, personalised learning, and support for Qur'anic learning. However, the evidence base remains emerging, and the contextual factors associated with GenAI implementation need to be distinguished according to their evidential source. Studies directly concerned with IRE or Islamic education indicate the importance of teacher preparedness, institutional readiness, cultural context, technological infrastructure, and ethical considerations [4, 28, 40]. Broader educational AI literature similarly identifies teacher competence, institutional support, infrastructure, governance, and AI literacy as important conditions for implementation [30]. Accordingly, these factors should not be treated as uniformly established IRE-specific findings; rather, they represent a combination of direct evidence from Islamic educational settings and contextual insights from the broader educational AI literature.

The importance of such distinctions becomes clearer because the consequences of inappropriate GenAI use in IRE may extend beyond conventional academic errors. Unchecked or insufficiently verified AI-generated content may contribute to misconceptions about religious concepts, inadequate evaluation of sources, and confusion between machine-generated information and knowledge conveyed by qualified teachers, scholars, or recognised religious sources. In disciplines where knowledge is also connected to moral and spiritual formation, the question is therefore not simply whether GenAI is useful or acceptable, but how its use can remain educationally responsible and consistent with the purposes and epistemic conditions of the discipline.

Accordingly, this review conceptualises ethical integration as the responsible implementation of GenAI in a pedagogically meaningful, transparent, and educator-monitored manner that addresses general AI ethics while also preserving the distinctive epistemic, moral, and religious requirements of IRE. General AI ethics remains relevant to IRE and includes concerns such as hallucination, bias, plagiarism, privacy, transparency, authorship, academic integrity, accountability, and excessive automation [4, 18, 24, 25, 36]. However, these concerns acquire additional significance in IRE because inaccurate, decontextualised, or unsourced information may affect learners' understanding of religious concepts, their evaluation of religious sources, and their perceptions of legitimate authority. Ethical integration in IRE therefore involves a layered approach in which general AI governance is supplemented by pedagogical safeguards and Islamic educational considerations, including religious accuracy, source authenticity, teacher mediation, and responsible engagement with religious knowledge.

Three conceptual lenses are particularly useful for articulating this Islamic dimension: *adab*, *amanah*, and epistemic authority. Their selection is not intended to suggest that they exhaust the range of Islamic ethical concepts relevant to GenAI. Rather, they were selected because they address three complementary dimensions of responsible engagement with knowledge. First, *adab* concerns the proper disposition and conduct of the learner towards knowledge, teachers, sources, and the learning process. This is consistent with Islamic educational scholarship that places *adab* and *ta'dib* at the centre of the educational process, linking intellectual development with appropriate conduct and the proper recognition of knowledge and its purposes [5]. Second, *amanah* concerns trust,

responsibility, and accountability. Within Islamic educational contexts, knowledge transmission itself can be understood as a responsibility that requires trustworthy handling and responsible practice. Contemporary Islamic education scholarship likewise connects *amanah* with the responsible acquisition and transmission of knowledge and with ethical conduct [1]. Third, epistemic authority concerns the legitimate basis for validating, interpreting, and relying upon knowledge claims. In IRE, this includes the roles of teachers, scholars, religious texts, and recognised educational institutions in guiding the interpretation and authentication of Islamic knowledge.

The distinction between religious authority and epistemic authority is important in this framework. Religious authority refers broadly to the recognised authority of religious persons, institutions, traditions, or sources in guiding religious belief, practice, interpretation, and communal understanding. Epistemic authority, by contrast, refers specifically to authority concerning knowledge: who or what is legitimately relied upon to determine, validate, interpret, or transmit knowledge claims. The two may overlap in IRE because teachers, scholars, texts, and institutions may possess both religious and epistemic functions; however, they are analytically distinct. This distinction is particularly relevant to GenAI because a system may be perceived by learners as an epistemically authoritative source simply because it produces fluent and apparently credible answers, without possessing the scholarly, textual, institutional, or religious legitimacy required for authoritative interpretation. Scholarship on religious authority similarly distinguishes authority associated with religious expertise and knowledge from authority understood primarily as power or command [10]. Thus, the concern in IRE is not only whether GenAI generates inaccurate information, but also whether its use changes learners' perceptions of who or what can legitimately be trusted as a source of religious knowledge.

Taken together, *adab*, *amanah*, and epistemic authority provide complementary analytical lenses for ethical integration. *Adab* addresses how learners engage with knowledge; *amanah* addresses how users act responsibly in producing, attributing, verifying, and sharing knowledge; and epistemic authority addresses how religious knowledge is validated and whose expertise is legitimately relied upon. These dimensions are particularly relevant to GenAI because the technology can simultaneously influence learner behaviour, responsibility for generated content, and perceptions of knowledge authority. They therefore provide a more targeted framework for IRE than applying general AI governance principles alone. The framework is consequently intended as a conceptual starting point for evaluating GenAI integration rather than as a claim that these three concepts constitute an exhaustive Islamic ethical framework.

The available literature on AI and education supports human-centred approaches to GenAI implementation involving privacy protection, sound pedagogical design, responsible policymaking, and institutional governance [30]. Similarly, literature concerning religious and Islamic education highlights teacher training, institutional support, culturally sensitive policy, verification of AI-generated information, and educator involvement in decision-making [4, 26, 28]. These considerations are particularly important in IRE because teachers are not merely facilitators of information but may also serve as moral educators and mediators of religious knowledge. Consequently, GenAI implementation in IRE requires a balance between technological usefulness, pedagogical value, religious authenticity, and moral responsibility.

Despite growing academic interest in AI and religion, the literature on GenAI in IRE remains limited, fragmented, and under-theorised. In this review, the term *under-theorised* refers specifically to the limited development of integrated conceptual frameworks that connect the pedagogical use of GenAI with Islamic ethical values, epistemological questions concerning knowledge and authority, and the theological and moral purposes of IRE. Existing studies frequently examine opportunities, risks, acceptability, pedagogy, AI ethics, Qur'anic learning, or responsible technology use separately, while relatively few studies systematically connect these dimensions within a coherent framework for IRE. The literature also contains limited classroom-based empirical evidence and few theoretical or evaluative frameworks capable of operationalising Islamic concepts such as *adab*, *amanah*, source authenticity, teacher mediation, and epistemic authority [6, 8, 36]. A systematic review is therefore warranted to synthesise the available evidence while distinguishing between direct IRE evidence and insights derived from adjacent fields. Accordingly, this systematic literature review aims to synthesise the emerging literature on the ethical integration of GenAI in IRE. The review is guided by the following research questions:

1. How is generative AI currently used in Islamic Religious Education?
2. What ethical concerns are reported regarding the use of generative AI in Islamic Religious Education?
3. What measures are recommended to support the ethical integration of generative AI in Islamic Religious Education?
4. What research gaps can be identified in the literature on generative AI in Islamic Religious Education?

Although this review is centred on IRE, the directly relevant literature remains limited. Therefore, adjacent studies from broader religious education and general educational AI ethics are included only when they contribute substantively to one or more of the review questions and address an issue that is conceptually relevant to IRE, such as religious knowledge and authority, teacher mediation, academic integrity, AI reliability, governance, or responsible use. Evidence from broader religious education is used primarily to illuminate issues that arise from the religious and theological character of the educational context, whereas general educational AI ethics literature is used to contextualise common technological and governance concerns. Such evidence is explicitly classified as indirect or contextual evidence and is not treated as direct evidence of GenAI implementation or educational effects in IRE. This distinction is intended to prevent inappropriate transfer of findings while allowing the limited IRE evidence base to be interpreted within the broader scholarship on religious education and educational AI ethics.

2 Literature review

2.1 Generative AI in education: opportunities and ethical concerns

The literature on generative artificial intelligence (GenAI) in the educational context has grown exponentially in recent years. Generally, the broader body of literature suggests that GenAI is a promising tool in education because of its ability to perform tasks such as content creation, feedback generation, assessment support, automated tutoring, adaptive learning, and personalised instruction. Prior research on artificial intelligence in higher education has indicated the rising popularity of AI applications utilised for

profiling, prediction, assessment, tutoring, and adaptive systems, although the pedagogical role of teachers in this line of research was rather weak [44, 45]. Recent studies on GenAI and large language models also demonstrate the potential of these technologies to help students and teachers explain ideas, write texts, provide formative feedback, generate ideas, and design instruction [24, 43].

At the same time, the literature shows that GenAI faces ethical and practical problems that cannot be ignored regarding its educational potential. Misinformation, hallucinations, plagiarism, bias, opacity, privacy risks, lack of accountability, authorship issues, and the undermining of academic integrity are among the main problems associated with GenAI in the educational context [6, 30, 43]. Moreover, some researchers argue that GenAI can contribute to superficial learning if students use machine-generated answers without thinking, evaluating sources and critically engaging with the information [24, 37]. Thus, the literature on GenAI in education is beginning to focus not on whether GenAI is useful but on how it should be integrated. Despite the important contribution of this ethical literature, however, it is too broad to describe specific problems of integrating GenAI in educational settings where knowledge has religious, moral and theological aspects.

2.2 Generative AI in Islamic religious education

From the emerging body of literature on the use of GenAI in Islamic Religious Education (IRE), both opportunities and potential risks have been identified. Existing studies generally position AI tools such as ChatGPT as pedagogical resources that can assist teachers rather than replace teachers, scholars, or recognised religious sources. Reported or proposed applications of GenAI in IRE include lesson planning, development of teaching materials, assessment preparation, student-centred learning, independent inquiry, classroom engagement, personalised learning, and teacher support [4, 8, 28, 40]. For example, Suhendi et al., [40] examine the potential uses of ChatGPT in designing IRE lessons, while Ahmad Taha et al., [4] examine the opportunities and challenges of ChatGPT in Islamic education through the lens of *maqasid al-shariah*.

Maqasid al-shariah refers broadly to the higher objectives of Islamic law and provides a normative basis for evaluating whether an action, practice, or technological application contributes to or threatens recognised human and religious interests. Classical formulations commonly identify the protection of religion (*hifz al-din*), life (*hifz al-nafs*), intellect (*hifz al-'aql*), lineage (*hifz al-nasl*), and property (*hifz al-mal*) as core objectives. In the context of GenAI, this perspective shifts the ethical question from whether a technology is merely permissible or useful to whether its purposes and consequences are compatible with these broader objectives. For example, the protection of religion may be relevant when GenAI is used to produce or interpret religious content; the protection of intellect may be relevant to concerns about cognitive dependency, superficial learning, and the uncritical acceptance of AI-generated answers; and the protection of property may be relevant to questions of authorship, intellectual property, and misuse of educational resources. Recent scholarship applying *maqasid al-shariah* to AI similarly identifies its potential to evaluate AI according to the preservation of faith, life, intellect, lineage, and wealth, while emphasising the need to balance technological benefits against possible harms.

However, *maqasid al-shariah* should not be treated in this review as a fully operational framework for evaluating every aspect of GenAI use in IRE. Rather, it provides an overarching normative orientation for asking whether GenAI use promotes legitimate educational and human benefits and avoids significant harms. The present review therefore distinguishes this broad normative level from more specific analytical concerns related to how knowledge is approached, how users act responsibly, and how religious knowledge is validated. This distinction is important because the literature on GenAI in IRE has not yet consistently operationalised *maqasid al-shariah* into classroom-level criteria for evaluating AI use.

Moreover, Majeed [28] found that Islamic teachers in Southern Pakistan perceive GenAI, including ChatGPT, as a potential pedagogical resource and an aid to teacher-student interaction. Similarly, Faizin et al., [16] provide evidence concerning Muslim learners' acceptance of AI in IRE. Such findings indicate that GenAI has potential pedagogical value and that users may be willing to adopt it. Nevertheless, technology acceptance and perceived usefulness should not be equated with educational effectiveness or ethical appropriateness. Acceptance studies primarily indicate whether learners or teachers are willing to use a technology; they do not establish whether GenAI improves learning outcomes, strengthens religious understanding, supports moral development, or produces responsible learner behaviour.

The current literature therefore requires empirical investigation beyond technology acceptance. In particular, future studies should examine whether GenAI affects learners' acquisition and retention of religious knowledge, their ability to distinguish authentic from unreliable religious sources, their critical evaluation of AI-generated religious information, their moral and ethical reasoning, and their capacity to engage independently with teachers and recognised religious sources. Classroom-based research should also investigate behavioural outcomes, including patterns of AI dependence, disclosure of AI use, academic integrity, source verification, and responsible use of AI-generated content. These dimensions are necessary because the educational value of GenAI in IRE cannot be adequately assessed through perceptions of usefulness or acceptance alone.

However, despite the potential uses of GenAI in IRE, the existing evidence remains limited. Much of the literature is conceptual, exploratory, or perception-based rather than based on observations of actual classroom implementation. Consequently, there is still insufficient empirical evidence concerning the effects of GenAI on learning processes, student behaviour, religious understanding, moral formation, and teacher-student relationships. This limitation is particularly important when interpreting findings from acceptance studies, as positive attitudes towards GenAI do not necessarily demonstrate educational effectiveness or ethical suitability.

2.3 Ethical integration, religious authority, and Islamic educational values

Ethical concerns are prominent in discussions of GenAI in IRE. In the broader educational literature, common concerns include plagiarism, cheating, unreliable outputs, lack of transparency, privacy violations, bias, authorship problems, and overreliance on AI systems [6, 25, 43]. These concerns are also relevant to IRE, but their implications may extend beyond conventional academic and technological risks because IRE involves religious knowledge, moral development, respect for knowledge, and the transmission and interpretation of religious teachings. Islamic education has consequently

been conceptualised as a holistic process encompassing intellectual, moral, spiritual, and social development [19, 39]. Ethical integration of GenAI in IRE therefore requires consideration not only of general AI governance but also of the epistemic and moral purposes of Islamic education.

At the broadest normative level, *maqasid al-shariah* provides a useful Islamic perspective for considering whether the purposes and consequences of GenAI use are consistent with the protection of fundamental religious and human interests. In IRE, *hifz al-din* is particularly relevant to the preservation of religious understanding and protection against doctrinal distortion, while *hifz al-‘aql* is relevant to the development and protection of learners’ intellectual capacities, including critical thinking and independent engagement with knowledge. Other objectives may become relevant depending on the application and context, including the protection of privacy, dignity, life, and property. Thus, *maqasid al-shariah* provides the review with a broad normative question: Does the use of GenAI in IRE contribute to legitimate educational and human benefit while avoiding harms that undermine the purposes of Islamic education?

Nevertheless, the *maqasid* perspective alone does not provide sufficient analytical specificity for the particular problems identified in the GenAI–IRE literature. It can establish broad objectives, but it does not by itself explain how learners should conduct themselves towards knowledge, how responsibility for AI-generated content should be exercised, or how the authority and authenticity of religious knowledge should be established. For this reason, the present review identifies *adab*, *amanah*, and epistemic authority as complementary analytical lenses rather than as competing alternatives to *maqasid al-shariah*. The relationship between these concepts is therefore hierarchical and complementary: *maqasid al-shariah* provides an overarching normative orientation, while *adab*, *amanah*, and epistemic authority provide more specific lenses for examining the conduct, responsibility, and knowledge-validation dimensions of GenAI use in IRE.

The selection of these three concepts is also not intended to suggest that they exhaust the Islamic ethical tradition or that concepts such as *ihsan*, *taqwa*, *sidq*, and *ikhlas* are irrelevant. Rather, they were selected because they directly correspond to three recurring problems identified in the literature on GenAI in IRE: the learner’s relationship with knowledge and learning, responsibility in the production and use of information, and the validation and authority of religious knowledge. *Adab* is therefore used to examine appropriate conduct towards knowledge, teachers, sources, and the learning process. *Amanah* is used to examine responsibility and trustworthiness in handling AI-generated information, authorship, student data, educational decisions, and religious content. Epistemic authority is used to examine how religious knowledge is validated, interpreted, and attributed to legitimate sources and knowledgeable persons. Together, these concepts provide a more focused analytical structure for examining the ethical integration of GenAI in IRE.

Within this framework, *adab* concerns the manner in which learners and educators engage with knowledge. It includes appropriate conduct towards teachers, sources, fellow learners, and the learning process. *Amanah* concerns responsibility and trustworthiness in the acquisition, production, attribution, verification, and dissemination of knowledge. In the context of GenAI, this includes disclosing AI assistance, verifying generated information, avoiding plagiarism, protecting student data, and accepting responsibility for the final use of AI-generated content. Epistemic authority, meanwhile,

concerns the legitimate basis on which knowledge claims are validated and interpreted. In IRE, this includes the roles of religious texts, teachers, scholars, and recognised educational institutions in authenticating and contextualising Islamic knowledge.

These concepts should therefore be understood primarily as analytical categories with normative foundations, rather than as a complete set of practical implementation rules. Their normative dimension derives from Islamic conceptions of proper conduct, responsibility, and legitimate knowledge, while their analytical function allows the literature to be examined systematically according to recurring ethical issues. The practical implications of these categories can subsequently be expressed through implementation criteria such as source verification, teacher mediation, responsible disclosure, academic-integrity procedures, protection of learner data, and restrictions on treating GenAI as an authoritative substitute for qualified religious guidance.

Under this approach, the ethical problem is not simply whether GenAI can produce inaccurate information, but also how its use may influence learners' relationships with knowledge, teachers, sources, and religious authority. Several studies raise concerns about AI accuracy, misuse of generated content, plagiarism, overreliance, privacy, and misalignment with Islamic values [4, 18, 25, 36]. In broader religious education, Papakostas [36], emphasises that AI should be examined not only from a pedagogical perspective but also in relation to theological interpretation, authority, and spiritual formation. Although such evidence does not constitute direct evidence of GenAI effects in IRE, it provides contextual insight into issues that may arise when AI is introduced into educational settings involving religious knowledge and formation. Similarly, studies of Islamic education emphasise the importance of teacher and institutional involvement in guiding AI use, validating generated material, and ensuring that technological innovations remain compatible with religious and cultural values [4, 26, 28].

Another important ethical issue is integrity. Research on GenAI in general education identifies plagiarism, assisted cheating, unclear authorship, and misuse of assessment as important ethical challenges [23, 25, 43]. These concerns are particularly relevant to the concept of *amanah* in IRE because academic integrity can also be understood as part of responsible moral conduct. If students use GenAI to generate religious explanations, assignments, or reflections without disclosure, verification, or meaningful personal engagement, the problem is simultaneously pedagogical and ethical. Ethical integration therefore requires policies concerning plagiarism, disclosure of AI use, assessment design, teacher guidance, and student education in responsible AI use.

Finally, readiness, policy, and institutional support remain important conditions for responsible GenAI implementation. Broader GenAI literature highlights teacher training, AI literacy, transparency, stakeholder involvement, and institutional governance [6, 30, 37]. Studies of Islamic education similarly indicate the importance of cultural readiness, infrastructure, teacher preparation, and institutional support [16, 28]. These findings reinforce the need for a multilayered approach in which individual users, teachers, institutions, and Islamic educational values jointly shape the responsible use of GenAI.

2.4 Gaps in the existing literature

Although the literature on GenAI in education is growing, several gaps remain regarding IRE. First, the evidence base is still empirically limited. Many studies discuss possible benefits, risks, and ethical considerations, but fewer studies examine actual classroom

implementation, student learning processes, teacher decision-making, or long-term educational effects in IRE settings. Second, the geographical and institutional scope of existing research remains narrow, with much of the directly relevant literature concentrated in specific national or educational contexts. Third, although Islamic values are frequently mentioned, concepts such as *adab*, *amanah*, religious authority, and epistemic authority are rarely operationalised as analytical categories for evaluating the use of GenAI in IRE. Fourth, existing reviews on GenAI ethics in education provide important background, but they do not sufficiently address the distinctive pedagogical, theological, epistemological, and cultural concerns of IRE [6, 8, 36].

These gaps highlight the need for a systematic literature review that specifically focuses on the ethical integration of GenAI in IRE. Such a review is necessary to distinguish between direct evidence from Islamic educational contexts and indirect insights drawn from broader religious education or general AI ethics literature. It is also needed to clarify how GenAI is currently discussed or used in IRE, what ethical concerns are reported, what safeguards are recommended, and what future research is required. In this study, *adab*, *amanah*, and epistemic authority are treated not merely as abstract Islamic concepts, but as conceptual lenses for interpreting ethical integration in relation to knowledge verification, teacher mediation, source authenticity, authorship, academic integrity, and responsible learner conduct.

3 Methodology

3.1 Review design

A systematic literature review (SLR) was conducted to synthesise the emerging literature on the ethical integration of generative artificial intelligence (GenAI) in Islamic Religious Education (IRE). An SLR was considered appropriate because research on GenAI in IRE remains relatively recent, fragmented, and methodologically diverse. Unlike a traditional narrative review, an SLR provides a structured process for identifying, screening, appraising, and synthesising relevant literature according to predefined procedures [29, 32, 35, 41]. This approach was particularly appropriate because the topic intersects several fields, including Islamic education, religious education, educational technology, generative AI, and AI ethics.

The review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement [35]. PRISMA was used to document the identification, screening, eligibility, and inclusion stages of the review. Searches were conducted in two multidisciplinary academic databases, Scopus and Web of Science. A total of 124 records were initially identified, comprising 78 records from Scopus and 46 from Web of Science. After the removal of 21 duplicate records and three records that did not meet the eligible publication-type requirements, 100 records remained for title and abstract screening.

To improve consistency in the screening process, study relevance was determined according to predefined eligibility criteria rather than a general judgement of whether a study was “sufficiently focused.” At the title and abstract screening stage, a record was considered potentially relevant when it addressed (a) Islamic Religious Education, Islamic education, religious education, Qur’anic education/studies, or a closely related religious-learning context; (b) generative AI, ChatGPT, large language models, or closely related generative AI technologies; and (c) at least one issue relevant to the review

questions, including educational use, ethical concerns, safeguards, governance, religious or epistemic implications, or identified research gaps. Records that did not satisfy these relevance conditions were excluded at this stage.

3.2 Identification of studies via databases and registers

Figure 1 above shows the PRISMA flow diagram of the study-selection process adapted from Page et al., [35]. During title and abstract screening, records were assessed against three predefined relevance conditions concerning the educational context, GenAI or related technologies, and relevance to the ethical, pedagogical, epistemic, governance, or research-gap dimensions addressed by the review.

3.3 Search strategy and database procedures

The literature search was conducted in Scopus and Web of Science, two multidisciplinary databases with broad coverage of education, social sciences, humanities, and technology-related research. These databases were selected to provide a consistent and reproducible search of peer-reviewed literature across the interdisciplinary scope of the review. Although databases such as ERIC, ATLA Religion Database, and ProQuest may contain additional education- or religion-specific studies, they were not included in this review. This represents a limitation of the search strategy, as relevant studies indexed exclusively in these databases may not have been captured.

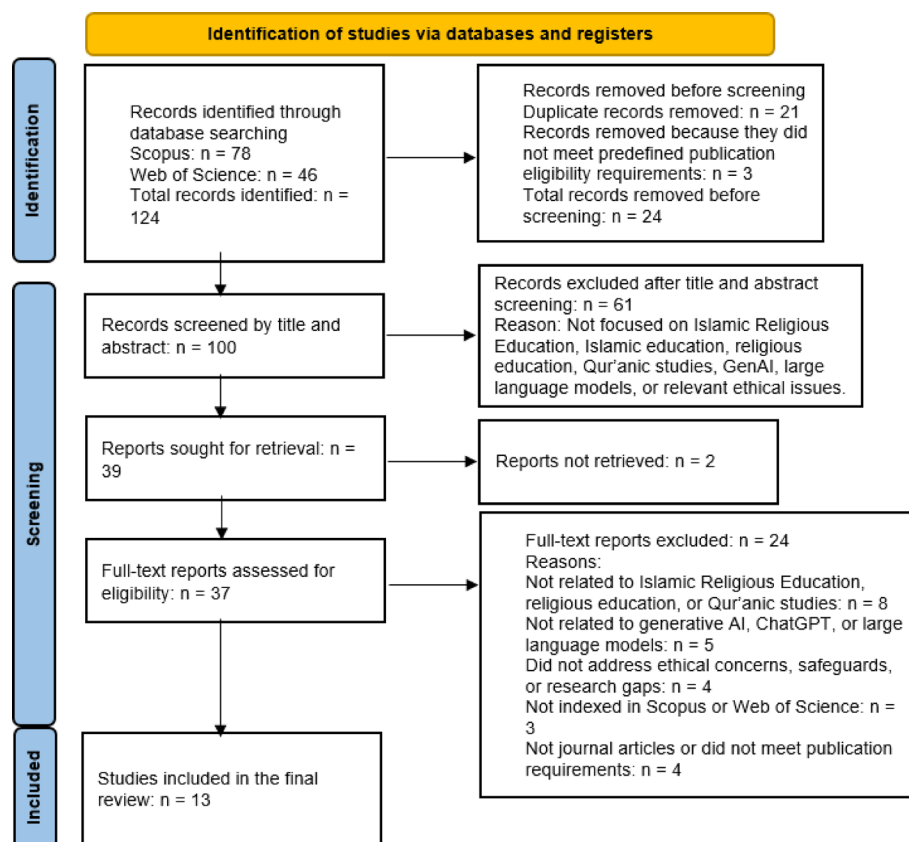


Fig. 1 PRISMA flow diagram of the study selection process adapted from Page et al., [35]

The search was conducted in April 2026 and covered the title, abstract, and keywords fields. The Boolean search strategy comprised three groups of terms: GenAI, Islamic/religious education, and ethical considerations. The main search string was: (“generative artificial intelligence” OR “generative AI” OR ChatGPT OR “large language model*” OR LLM*) AND (“Islamic Religious Education” OR “Islamic education” OR “religious education” OR “Qur’anic studies” OR “Quranic learning”) AND (ethic* OR “academic integrity” OR plagiarism OR privacy OR bias OR transparency OR governance OR “responsible integration” OR “AI literacy”). The search was limited to peer-reviewed English-language journal articles published between 2022 and 2026 and indexed in Scopus or Web of Science. 2022 was selected as the starting point because the public introduction of ChatGPT in November 2022 marked a major transition in the widespread availability and educational visibility of contemporary generative AI. Records were exported, duplicates removed, and the remaining records screened according to the pre-defined eligibility criteria and PRISMA 2020 guidelines [35]. The initial search yielded 124 records, comprising 78 from Scopus and 46 from Web of Science.

3.4 Inclusion and exclusion criteria

The inclusion and exclusion criteria were established prior to the full-text eligibility evaluation to maintain consistency, transparency, and alignment with the research objectives. The inclusion criteria were as follows: (a) the study concerned generative AI, ChatGPT, large language models, or other AI systems relevant to the research questions; (b) the study addressed IRE, Islamic education, Qur’anic studies, religious education, or educational AI ethics and demonstrated a substantive connection to IRE; specifically, a study was considered relevant when its findings or discussion addressed the use of GenAI in Islamic or religious learning, the teaching or learning of Islamic/religious knowledge, or ethical, pedagogical, epistemic, or governance issues applicable to GenAI use in IRE; (c) at least one dimension of ethical integration, including academic integrity, plagiarism, privacy, bias, transparency, governance, value alignment, teacher supervision, religious authority, safeguards, or research gaps, was considered; (d) the paper was published as a peer-reviewed journal article indexed in Scopus or Web of Science; (e) the article was written in English; and (f) the full text was obtainable.

The exclusion criteria comprised: (a) studies concerning educational technology without substantive relevance to GenAI or large language models; (b) studies with no substantive connection to education, religious learning, or IRE; (c) studies that did not address any ethical, pedagogical, epistemic, or governance dimension relevant to the review questions; (d) studies not indexed in Scopus or Web of Science; (e) studies for which the full text was unavailable; (f) non-peer-reviewed publications; and (g) publications in languages other than English. These criteria were applied consistently during title/abstract screening and full-text eligibility assessment in accordance with the PRISMA 2020 reporting framework [35].

3.5 Study screening and selection process

The screening and selection were conducted step by step in accordance with the PRISMA 2020 protocol [35]. After 21 duplicate records and three records that did not meet the predefined publication eligibility requirements were removed, 100 records remained for title and abstract screening. During title and abstract screening, 61 records

were excluded because they did not meet the predefined relevance criteria concerning the educational context, generative AI or related technologies, and issues relevant to the review questions. The remaining 39 reports were sought for full-text retrieval. Two reports could not be retrieved because their full texts were not accessible through the available institutional subscriptions. Consequently, 37 full-text reports were assessed against the predefined inclusion and exclusion criteria.

The title and abstract screening and subsequent full-text eligibility assessment were conducted independently by two reviewers using the predefined inclusion and exclusion criteria. Screening decisions were compared after each stage, and disagreements were resolved through discussion and consensus. Where necessary, unresolved cases were discussed with the wider research team before a final inclusion decision was made. A formal inter-reviewer agreement statistic was not calculated. However, the use of predefined eligibility criteria and consensus procedures was intended to enhance consistency in study selection. This process resulted in the inclusion of 13 studies for the final synthesis.

3.6 Data extraction and organisation

Information was extracted from the 13 included studies using a structured review matrix developed in accordance with the research questions. The matrix captured the author and year, title, database indexing, location or educational context, methodology, participants or data source, evidence type, relevance to IRE, key findings, ethical issues, recommendations, and research gaps. The extracted information was organised according to the four review questions concerning: (1) current applications of GenAI in IRE; (2) ethical and epistemic concerns; (3) measures for responsible GenAI integration; and (4) research gaps. The review matrix was used to facilitate systematic comparison across studies with different methodological designs, including empirical, conceptual, review, and broader AI ethics studies. The complete data-extraction matrix is provided as Supplementary Material S1 to enhance the transparency and reproducibility of the review process. Table 2 presents a condensed summary of the methodological characteristics, evidence categories, and thematic contributions of the included studies. The matrix also enabled the researchers to distinguish between direct evidence from IRE contexts and contextual evidence drawn from broader religious education and educational AI ethics literature.

3.7 Classification of direct, indirect, and contextual evidence

Because directly relevant studies on GenAI in IRE remain limited, the review distinguished among direct IRE evidence, indirect religious education evidence, and contextual GenAI evidence. These categories were applied during data extraction and synthesis and were not treated as equivalent forms of evidence. Direct IRE evidence referred to studies focused specifically on IRE, Islamic education, Qur'anic learning, Islamic teachers, or Muslim learners. Indirect religious education evidence referred to studies examining AI within broader religious or theological education contexts without an Islamic-specific focus. Contextual GenAI evidence comprised studies addressing broader educational GenAI issues such as academic integrity, AI literacy, governance, responsible use, reliability, and educational AI ethics without directly examining religious education.

The distinction was applied according to the context of the evidence and its relevance to the review questions. Where direct IRE evidence was available, it was used to substantiate IRE-specific findings. Where direct evidence was limited or unavailable, indirect evidence was used only to provide contextual support or identify issues that may be relevant to IRE; such evidence was not used to claim that a finding had been empirically established in IRE. In the thematic synthesis, the source context of each finding was therefore retained to distinguish directly reported IRE evidence from contextual evidence inferred from adjacent literature. This procedure was intended to minimise inappropriate generalisation from broader educational or religious education research to IRE-specific contexts.

3.8 Quality appraisal of included studies

The methodological and reporting quality of the included studies as shown in Table 1 was assessed using an appraisal framework adapted from Abouzahra et al., [2]. This framework was selected because the review included heterogeneous evidence, comprising quantitative surveys, qualitative studies, mixed-methods research, conceptual papers, literature-based studies, technical evaluations, and application reviews. Design-specific tools such as CASP and JBI are available for particular methodological designs, while MMAT is designed primarily for the appraisal of empirical qualitative, quantitative, and mixed-methods studies. Because the present synthesis also included conceptual and literature-based evidence, the adapted Abouzahra et al., [2] framework provided a common set of criteria for assessing clarity, methodological transparency, conceptual adequacy, and reporting quality across different types of studies. This approach has also been adopted in subsequent systematic literature reviews across different applied fields [14, 33].

Six quality assessment criteria were applied. QA1 assessed whether the purpose of the study was clearly stated. QA2 assessed whether the relevance and usefulness of the work were clearly presented. QA3 assessed whether the methodology was clearly established. QA4 examined whether the key concepts, constructs, or theoretical ideas were clearly defined. QA5 assessed whether the study was appropriately positioned in relation to previous work. QA6 examined whether the study's limitations were clearly reported. Each criterion was scored as Yes = 1, Partly = 0.5, or No = 0, giving a maximum score of 6.0. To promote consistency in scoring, the same operational definitions were applied across all studies, with "Yes" indicating that the criterion was clearly and sufficiently addressed, "Partly" indicating that it was addressed but incompletely or with limitations, and "No" indicating that the criterion was not adequately addressed.

Although the same six appraisal domains were applied across all included studies, the criteria were interpreted in relation to the methodological and evidential characteristics of each study design. For empirical studies, appraisal focused on the clarity and adequacy of the research design, participants or data sources, procedures, and limitations. For conceptual, literature-based, and technical studies, the criteria were applied to the clarity of the analytical approach, conceptual development, evidential basis, positioning within previous scholarship, and acknowledgement of limitations where applicable. Consistent with the purpose of the adapted appraisal framework [2], the criteria were therefore used as a common assessment of reporting, methodological transparency, and conceptual adequacy rather than as a design-specific risk-of-bias instrument.

Table 1 Quality appraisal of the 13 included studies

No.	Study	QA1 purpose	QA2 usefulness	QA3 methodology	QA4 concepts	QA5 comparison	QA6 limi- ta- tions	Total score	De- ci- sion
1	Faizin et al., [16]	1	1	1	1	1	1	6.0	In- clud- ed
2	Suhendi et al., [40]	1	1	1	1	0.5	0	4.5	In- clud- ed
3	Achruh et al., [3]	1	1	1	1	0.5	1	5.5	In- clud- ed
4	Majeed [28],	1	1	1	1	1	1	6.0	In- clud- ed
5	Birgün [11],	1	1	1	1	1	1	6.0	In- clud- ed
6	Papakostas [36],	1	1	0.5	1	1	0	4.5	In- clud- ed
7	Alsharefeen and Al Sayari [7],	1	1	1	1	1	1	6.0	In- clud- ed
8	Cotton et al., [13]	1	1	0.5	1	1	0	4.5	In- clud- ed
9	Chan [12],	1	1	1	1	1	1	6.0	In- clud- ed
10	Mogavi et al., [31]	1	1	1	1	1	1	6.0	In- clud- ed
11	Ravšelj et al., [38]	1	1	1	1	1	1	6.0	In- clud- ed
12	Kurata et al., [26]	1	1	1	1	1	1	6.0	In- clud- ed
13	Day [15],	1	1	0.5	1	0.5	0	4.0	In- clud- ed

Accordingly, the resulting scores were not interpreted as indicating methodological equivalence across different types of evidence.

The quality appraisal was conducted independently by two reviewers using the six pre-defined assessment criteria. Each reviewer evaluated and scored the 13 included studies separately before the appraisal results were compared. Differences in scoring were discussed with reference to the predefined operational definitions for each criterion and resolved through consensus. Where necessary, the wider research team was consulted before a final score was assigned. A formal inter-reviewer agreement statistic was not calculated for the quality appraisal.

A score of 3.0/6.0 was used as a descriptive adequacy threshold, representing 50% of the maximum possible score. This threshold was adopted as a pragmatic criterion for identifying studies with a minimum level of reporting and methodological clarity rather

than as a validated cut-off for methodological quality or risk of bias. Given the emerging and heterogeneous nature of the literature, appraisal scores were not used as the sole basis for study exclusion. Instead, they were considered together with study design, evidence type, and contextual relevance when interpreting the synthesis.

The appraisal results indicated that all 13 included studies met the predefined descriptive adequacy threshold. Total scores ranged from 4.0 to 6.0, with eight studies achieving 6.0, one scoring 5.5, three scoring 4.5, and one scoring 4.0. No study scored below the threshold of 3.0. The relatively narrow distribution of scores suggests that the appraisal framework had limited capacity to discriminate fine-grained differences in reporting and methodological characteristics across the heterogeneous study designs included in the review. This may partly reflect the broad purpose of the framework, which was to assess reporting clarity, methodological transparency, conceptual adequacy, and acknowledgment of limitations rather than to provide a design-specific assessment of methodological rigour or risk of bias. The scores were therefore not interpreted as demonstrating equivalence in methodological quality across the included studies.

The quality appraisal was used to inform the interpretation and evidential contribution of the included studies rather than to determine inclusion solely on the basis of a numerical cut-off. This approach is consistent with systematic review guidance indicating that critical appraisal should inform the synthesis and interpretation of evidence while taking account of the methodological characteristics of included studies (Aromataris et al., [9]; [21]). Studies with clearer methodological procedures and stronger empirical foundations were therefore used more confidently when supporting claims concerning reported learner and teacher experiences, perceptions, and educational practices. In contrast, conceptual, literature-based, and technical studies were used primarily to identify theoretical, ethical, epistemological, and pedagogical issues and were not treated as equivalent evidence of observed classroom effects. This distinction is consistent with the principle that methodological appraisal should recognise the differing characteristics and evidential contributions of heterogeneous study designs [21, 22]. Lower-scoring studies were retained because they remained relevant to the review questions, but their findings were interpreted more cautiously and were not used as the sole basis for substantive conclusions. The appraisal therefore functioned as a means of calibrating the strength of claims made in the synthesis rather than as a mechanism for numerical weighting or automatic study exclusion.

Contextual relevance was also considered during synthesis. Direct evidence from IRE, Islamic education, Muslim learners, Islamic teachers, and Qur'anic learning was used to support IRE-specific claims, whereas evidence from broader religious education and general GenAI literature was used only as contextual or supporting evidence. This distinction helped prevent findings from adjacent fields from being interpreted as direct evidence of GenAI implementation or effects in IRE.

Overall, the evidence base remains uneven. Direct research on GenAI in IRE is limited and includes a substantial proportion of perception-based, conceptual, exploratory, and context-specific studies. Broader GenAI literature provides useful contextual evidence concerning academic integrity, learner perceptions, policy, governance, and responsible AI use, but these findings cannot be assumed to represent IRE settings. The findings of

this review should therefore be interpreted as an emerging synthesis of the available evidence rather than as conclusive evidence of the classroom effects of GenAI in IRE.

3.9 Data synthesis and thematic analysis

The included studies were analysed through thematic analysis as the body of evidence was made up of different types of studies, such as conceptual studies, empirical research, literature reviews, and more broadly, AI ethics studies. Synthesis involved both deductive and inductive coding. Codes were developed using ATLAS.ti, which was employed to manage extracted data, generate codes, compare coded units, and help form thematic categories. The initial deductive codes were created based on the four research questions and related to GenAI application, ethical issues, responsible integration actions, and research gaps. Inductive coding involved rereading all included studies to identify recurring concerns and patterns, including doctrinal accuracy, source authenticity, responsibility for AI-generated content, teacher supervision, academic integrity, privacy, bias, religious authority, value alignment, and AI literacy. These recurring concerns were subsequently interpreted through the complementary analytical lenses of *adab*, *amanah*, and epistemic authority. The coded data were compared and grouped into categories, which were then clustered into four final themes: pedagogical uses of GenAI in IRE; ethical and epistemic risks associated with the use of GenAI in IRE; safeguards for the responsible integration of GenAI in IRE; and research gaps and future directions. Such an approach has helped achieve a balance between synthesising diverse types of evidence and differentiating between direct and indirect evidence in the context of IRE and educational AI ethics [17, 42].

3.10 Trustworthiness of the review

Several techniques were used to enhance the credibility of the review. First, the review adopted the PRISMA 2020 approach, which supports transparent reporting of how studies were identified, screened, selected, and included [35]. Second, the inclusion and exclusion criteria were defined prior to full-text screening to enhance consistency in study selection. Trustworthiness was further strengthened through independent screening and quality appraisal by two reviewers, comparison of decisions and scores, and resolution of disagreements through consensus, with wider-team consultation where necessary. Third, a structured data-extraction matrix was used to organise information on context, design, data source, evidence type, relevance to IRE, ethical issues, recommended measures, and research gaps. Finally, the classification of direct IRE, indirect religious education, and contextual GenAI evidence helped distinguish IRE-specific findings from evidence derived from adjacent literature.

4 Results

Table 2 summarises the methodological characteristics, evidence type, contextual relevance, and thematic contribution of the 13 included studies. The studies employed diverse methodological approaches, including quantitative, qualitative, and mixed-methods research, as well as conceptual analyses, literature reviews, and technical evaluations. This methodological heterogeneity reflects the emerging and multidisciplinary nature of research at the intersection of GenAI, Islamic Religious Education (IRE), religious education, academic integrity, and educational AI ethics. To support a consistent

synthesis, studies were classified according to their contextual relevance to the review questions and their degree of direct engagement with IRE.

Five studies were classified as direct IRE/Islamic education evidence because they addressed IRE, Islamic education, Islamic teachers, Muslim learners, or Qur'anic learning [3, 11, 16, 28, 40]. Two studies were classified as indirect religious education evidence because they examined AI in broader religious education or theological contexts without focusing specifically on Islamic education [26, 36]. The remaining six studies were classified as contextual GenAI evidence because they addressed GenAI use, academic integrity, learner attitudes, AI reliability, or educational AI ethics without examining IRE directly [7, 12, 13, 15, 31, 38].

The classification was based on predefined criteria established during the data-extraction process. Direct evidence was identified when the study population, educational setting, or primary research focus was explicitly situated in IRE or a closely related Islamic educational context. Indirect religious education evidence referred to studies addressing religious or theological education without an Islamic-specific focus, while contextual GenAI evidence addressed broader educational AI or GenAI issues without a specific religious-education context. These categories determined how findings were used in the synthesis rather than assigning numerical weights to individual studies. Direct evidence was used to substantiate claims about GenAI in IRE, whereas indirect and contextual evidence was used to contextualise or supplement the interpretation of ethical, pedagogical, epistemological, and governance issues. Findings from indirect or contextual studies were not treated as empirical evidence of GenAI effects in IRE.

The included studies were not treated as providing equivalent forms of evidence. Empirical studies were used primarily to support claims concerning reported experiences, perceptions, and educational practices, whereas conceptual, literature-based, and technical studies primarily informed theoretical, ethical, epistemological, and applied interpretation.

4.1 Thematic analysis and synthesis

The four themes were developed through a combination of deductive and inductive thematic analysis. Deductive analysis involved organising the initial codes according to the four research questions: (1) current uses of GenAI; (2) ethical implications of GenAI; (3) strategies for responsible integration; and (4) research gaps in GenAI and IRE. Inductive analysis was then used to identify recurring patterns across the included studies, including lesson planning, assessment, personalisation, academic integrity, privacy, hallucinations, doctrinal accuracy, *adab*, *amanah*, epistemic authority, teacher supervision, value alignment, and AI literacy. Related codes were subsequently compared and grouped into four overarching themes: (1) applications of GenAI in IRE pedagogy; (2) ethical, epistemic, and value-alignment risks; (3) responsible integration of GenAI; and (4) research gaps and future directions. During thematic interpretation, the contextual relevance of each study was retained as an explicit criterion. Findings from IRE and Islamic education studies were used to substantiate claims specifically concerning GenAI in IRE, while findings from broader religious education and contextual GenAI studies were used to provide supporting or contextual evidence. Where evidence from broader literature was incorporated into a theme, it was not treated as direct evidence of GenAI use or effects

Table 2 Methodological characteristics, evidence category, and thematic mapping of included studies

Author & year	Title	Database	Country / context	Design / participants / data source	Evidence type	Relevance category	Themes mapped	Evidential role in the synthesis
Faizin et al., [16]	<i>Muslim students' acceptance of artificial intelligence in Islamic religious education: An extended TAM approach</i>	Scopus	Indonesia; IRE in higher education	Quantitative survey; 224 Muslim university students; PLS-SEM	Empirical–quantitative, perception-based	Direct IRE	1, 2, 3, 4	Direct empirical support for student acceptance of AI in IRE and reported concerns regarding credibility, plagiarism, and value alignment.
Suhendi et al., [40]	<i>The emergence of ChatGPT: Exploring its potential in designing Islamic religious education learning</i>	Scopus	Indonesia; IRE learning design	Library / qualitative literature study	Conceptual / literature-based	Direct IRE	1, 2, 3, 4	Direct IRE literature-based support for proposed uses in lesson planning, teaching materials, assessment, and administration, together with concerns about accuracy and misuse.
Achruh et al., [3]	<i>Challenges and Opportunities of Artificial Intelligence Adoption in Islamic Education in Indonesian Higher Education Institutions</i>	Scopus	Indonesia; Islamic higher education	Qualitative interviews with scholars	Empirical–qualitative	Direct IRE	1, 2, 3, 4	Direct empirical support for reported opportunities and challenges of AI adoption in Islamic education, including personalisation, efficiency, privacy, bias, transparency, and institutional readiness.
Majeed [28],	<i>The role of ChatGPT and generative AI in shaping the future of Islamic teachers' pedagogy in Southern Pakistan</i>	Scopus	Southern Pakistan; Islamic secondary schools	Quantitative questionnaire; 440 teachers; SPSS	Empirical–quantitative	Direct IRE	1, 2, 3, 4	Direct empirical support for teacher adoption and perceptions of GenAI, including the importance of cultural fit, infrastructure, and teacher training in Islamic educational settings.
Birgün [11],	<i>Integrating AI into Quran Learning Technical Advances and Pedagogical Gaps</i>	Scopus	Quranic learning / religious learning	Qualitative case study + application review	Empirical–qualitative / applied	Direct IRE	1, 2, 3, 4	Direct applied and qualitative support for potential AI-assisted tajweed, pronunciation, and adaptive-feedback functions while emphasising continued teacher-mediated Quranic learning.
Kurata et al., [26]	<i>Teaching religious studies with artificial intelligence: A qualitative analysis of Lesotho secondary schools teachers' perceptions</i>	Scopus/WoS	Lesotho; secondary religious studies	Qualitative interviews; 20 teachers	Empirical–qualitative	Indirect religious education	1, 2, 3, 4	Contextual empirical support from broader religious education concerning teacher perceptions of efficiency, engagement, access barriers, ethical issues, and policy and training needs.
Papakostas [36],	<i>Artificial Intelligence in Religious Education: Ethical, Pedagogical, and Theological Perspectives</i>	Scopus/WoS	Broader religious education / theology	Conceptual analysis	Conceptual	Indirect religious education	1, 2, 3, 4	Conceptual contextual support for interpreting theological distortion, shifts in religious authority, privacy, bias, and the need for educator oversight in religious education.

Table 2 (continued)

Author & year	Title	Database	Country / context	Design / participants / data source	Evidence type	Relevance category	Themes mapped	Evidential role in the synthesis
Alshareef and Al Sayari [7],	<i>Examining academic integrity policy and practice in the era of AI: A case study of faculty perspectives</i>	Scopus/WoS	UAE higher education	Mixed methods; 71 survey responses + 17 interviews	Empirical–mixed methods	Contextual GenAI	2, 3, 4	Contextual empirical support for academic-integrity policy and governance issues, including policy ambiguity, detection limitations, workload, and culturally sensitive institutional responses.
Cotton et al., [13]	<i>Chatting and cheating: Ensuring academic integrity in the era of ChatGPT</i>	Scopus	Higher education	Conceptual / critical analysis	Conceptual	Contextual GenAI	2, 3, 4	Conceptual contextual support for interpreting AI-enabled cheating, authorship concerns, academic integrity, and the need for assessment redesign.
Chan [12],	<i>Students' perceptions of "AI-giarism": Investigating changes in understandings of academic misconduct</i>	Scopus/WoS	Higher education	Empirical survey; undergraduate and postgraduate students	Empirical–quantitative	Contextual GenAI	2, 3, 4	Contextual empirical support for students' differing understandings of AI-assisted academic misconduct and the need for clearer institutional guidance and policy.
Mogavi et al., [31]	<i>ChatGPT in education: A blessing or a curse? A qualitative study exploring early adopters' utilization and perceptions</i>	Scopus/WoS	General education; early adopters	Qualitative content analysis of online user discussions	Empirical–qualitative	Contextual GenAI	1, 2, 4	Contextual empirical support for perceived productivity and learning-support benefits alongside risks of overreliance and superficial learning among early GenAI users.
Ravšelj et al., [38]	<i>Higher education students' perceptions of ChatGPT: A global study of early reactions</i>	Scopus/WoS	Global higher education	Empirical survey; higher education students	Empirical–quantitative	Contextual GenAI	1, 2, 3, 4	Contextual empirical support for broad student perceptions of ChatGPT usefulness and limitations and for the need for responsible-use guidance.
Day [15],	<i>A preliminary investigation of fake peer-reviewed citations and references generated by ChatGPT</i>	Scopus	Higher education / academic writing	Empirical test of ChatGPT-generated references	Empirical–technical evaluation	Contextual GenAI	2, 3, 4	Contextual technical evidence demonstrating fabricated or inaccurate AI-generated citations and supporting verification and source-authenticity safeguards.

in IRE. This procedure was applied consistently across all four themes to distinguish IRE-specific findings from insights derived from adjacent literature.

4.1.1 Theme 1: pedagogical applications of GenAI in IRE

The first theme concerns the potential pedagogical applications of GenAI in IRE. Evidence from direct IRE and Islamic education studies suggests that GenAI is currently viewed primarily as an auxiliary tool rather than a substitute for teachers. Reported or proposed applications include lesson planning, teaching-material preparation, assessment development, administrative support, personalised learning, adaptive learning, student-centred activities, and teacher–student interaction [3, 16, 28, 40]. In Qur’anic learning, GenAI has also been explored for applications such as pronunciation assessment, *tajweed* recognition, and adaptive feedback, although these applications remain supplementary to teacher-mediated instruction [11]. Similar possibilities have been identified in broader religious education, where AI may support teaching efficiency, student engagement, and learning processes [26, 36]. Contextual GenAI studies likewise suggest potential benefits for productivity, explanation, learning support, and motivation [31, 38]. However, these findings should be interpreted as emerging possibilities rather than established evidence of pedagogical effectiveness in IRE. Much of the available literature is exploratory, conceptual, or perception-based, and there is limited empirical evidence from actual IRE classroom implementation. Therefore, the current evidence supports the potential usefulness and proposed applications of GenAI in IRE, but does not yet establish their effects on learning outcomes, religious understanding, student behaviour, or teacher–student relationships.

4.1.2 Theme 2: ethical, epistemic, and value-alignment risks

The second theme concerns the ethical, epistemic, and value-alignment risks associated with GenAI in IRE. Some concerns identified in the direct IRE literature, including plagiarism, cheating, unreliable outputs, privacy, bias, transparency, and AI readiness, are shared with the broader educational AI literature [3, 16, 28, 40]. Contextual studies similarly identify academic integrity, fabricated citations, hallucinations, and over-reliance on AI as broader GenAI-related concerns [7, 12, 13, 15, 31]. However, these concerns may acquire additional significance in IRE because GenAI is used in relation to religious knowledge, theological interpretation, and moral formation. Direct and indirect religious education studies highlight concerns regarding the accuracy and appropriate use of religious information, source authenticity, religious authority, spiritual formation, and potential misalignment with Islamic values [11, 36]. Thus, while issues such as plagiarism, privacy, and bias are not unique to IRE, inaccurate or fabricated information may have particular implications when it concerns Qur’anic knowledge, religious beliefs, or moral guidance. These concerns should nevertheless be understood as potential risks rather than established consequences of GenAI use in IRE, given the limited direct empirical evidence.

4.1.3 Theme 3: safeguards for responsible integration

Theme three focuses on the precautions that should be put in place to ensure responsible GenAI implementation. The direct IRE and Islamic education literature consistently emphasises the importance of teacher oversight, verification of AI outputs, alignment

with Islamic values, digital competence, cultural appropriateness, institutional support, and teacher training [3, 16, 28, 40]. The safeguards in Qur'anic education include ensuring that AI does not replace conventional instruction, especially when there is an emphasis on recitation, tajweed, the relational and spiritual transmission of Qur'anic knowledge, and teacher guidance [11]. Precautions suggested through indirect religious education literature also include oversight by educators, theological-ethical guidance, policy support, infrastructural enhancement, and educator training [26, 36]. The GenAI literature across contexts reinforces the need for academic integrity policies, the rethinking of assessment, guidance for students, AI literacy, responsible disclosure procedures, and verification of AI-generated sources [7, 12, 13, 15]. In general, these studies indicate that responsible implementation necessitates multi-layered precautions which include teachers, students, institutions, and ethical standards rather than individual users' personal judgment alone.

4.1.4 Theme 4: research gaps and future directions

The fourth major theme addresses the limited and evolving nature of the literature. Research on the integration of GenAI into IRE and Islamic education tends to focus on perceived usefulness, acceptability, conceptual opportunities, and general ethical issues rather than classroom implementation [16, 28, 40]. While the qualitative and applied literature provides valuable insights, there is still a need for classroom-based, longitudinal, comparative, and intervention studies examining the effects of GenAI on teaching, learning, religious knowledge, academic integrity, and teacher–student relationships in IRE [3, 11]. Indirect religious education literature further indicates the need for discipline-specific research on theological guidance, religious authority, and educator mediation [26, 36]. Contextual GenAI literature identifies additional gaps concerning student AI literacy, academic integrity policies, responsible-use guidelines, and verification of generated content [7, 12, 13, 15, 38].

Hence, future research should move beyond general discussions of opportunities and risks towards empirically grounded frameworks for GenAI implementation in IRE. The proposed emphasis on *adab*, *amanah*, source authenticity, teacher mediation, and epistemic authority follows directly from the gaps identified in the reviewed literature: these concepts address learners' conduct towards knowledge, responsibility in using AI-generated information, the authenticity of religious sources, the role of educators, and the legitimate validation of religious knowledge. They therefore complement rather than replace broader AI-ethics frameworks, which primarily address concerns such as privacy, transparency, fairness, accountability, and academic integrity. Further research should examine how these Islamic educational principles can be operationalised and empirically evaluated in GenAI-supported IRE settings.

5 Discussion

5.1 GenAI as a bounded pedagogical assistant in IRE

The findings suggest that GenAI is best understood as a bounded pedagogical assistant in IRE rather than as a replacement for teachers, scholars, or recognised religious sources. The literature primarily associates GenAI with practical pedagogical activities such as lesson planning, teaching-material development, assessment preparation, administrative support, personalised learning, and teacher–student interaction [3, 16,

28, 40]. Regarding Qur'anic learning, the possible uses of AI may be confined to its technical abilities to analyse pronunciation or detect *tajweed*; however, they would be limited compared with the spiritual, relational, and pedagogical importance of traditional instruction [11]. Such results correspond to the general picture drawn by the GenAI education literature, as the latter acknowledges the usefulness of GenAI for explanation, writing assistance, provision of feedback and learning assistance, but emphasises human supervision and critical evaluation [24, 31, 38].

The bounded role of GenAI is particularly important in IRE because religious education involves not only the transmission of knowledge but also moral formation, interpretation, respect for knowledge, and the cultivation of religious understanding. GenAI may therefore assist with specific pedagogical tasks, but the available evidence does not support its substitution for the religious educator, moral guide, or legitimate source of epistemic authority. There is currently no robust evidence that GenAI has had a transformative impact on IRE classrooms. Instead, the literature demonstrates perceived usefulness alongside limited evidence of actual pedagogical effects [16, 28].

5.2 Ethical integration beyond general AI governance

The literature review indicates that ethical integration of GenAI in IRE extends beyond general AI governance. Concerns such as privacy, transparency, bias, hallucinations, accountability, academic integrity, authorship, and plagiarism remain relevant to IRE [6, 7, 12, 13, 25, 43]. However, these concerns are not unique to IRE. The distinctive issue is how such risks interact with the particular epistemic and educational purposes of Islamic education. In IRE, inaccurate, decontextualised, or unsourced AI-generated content may affect not only academic performance but also learners' understanding of religious concepts, perceptions of source authenticity, and trust in legitimate religious authorities. Direct IRE research identifies concerns regarding plagiarism, misuse of sources, unreliable outputs, privacy, bias, cultural readiness, and alignment with Islamic values [3, 16, 40], while broader religious education literature raises concerns about theological interpretation, religious authority, and spiritual formation [36]. These latter concerns remain potential rather than empirically established consequences in IRE.

Accordingly, ethical integration in IRE can be understood at three complementary levels. At the general AI-ethics level, it involves privacy, transparency, accountability, bias awareness, and academic integrity. At the pedagogical level, it involves teacher supervision, critical evaluation of AI outputs, assessment redesign, AI literacy, and responsible disclosure. At the Islamic educational level, it additionally requires attention to religious accuracy, *adab*, *amanah*, source authenticity, teacher mediation, and epistemic authority. These principles have practical implications: *adab* may require learners to engage respectfully and critically with religious knowledge rather than treating AI as an unquestioned authority; *amanah* may require disclosure of AI use, verification of generated religious content, and responsible handling of data and authorship; source authenticity may require checking AI-generated religious claims against recognised texts and reliable sources; and teacher mediation may require educators to verify, contextualise, and correct AI-generated religious information. Epistemic authority therefore remains with recognised religious sources, qualified teachers, and scholars rather than with the GenAI system itself. The distinctive contribution of the Islamic perspective is therefore not to replace general AI governance, but to extend it by addressing how GenAI should

be situated within the epistemic, moral, and pedagogical purposes of Islamic education. This perspective shifts the discussion from whether GenAI is simply useful or permissible towards how it can be used responsibly without compromising the authenticity of religious knowledge, the role of educators, and the moral purposes of IRE.

5.3 Toward an Islamic ethical framework for GenAI integration

The findings of this systematic review suggest the need for an Islamic ethical framework for GenAI integration in IRE. However, the framework proposed here is not derived by selecting Islamic concepts *a priori*. Rather, it emerged from recurring concerns identified across the reviewed literature concerning the responsible use of knowledge, responsibility for AI-generated content, the authenticity and validation of religious knowledge, the role of teachers, and alignment between technology and the purposes of Islamic education. These recurring concerns were subsequently mapped onto relevant Islamic educational concepts. On this basis, *adab*, *amanah*, and epistemic authority are proposed as core analytical lenses, while source authenticity, teacher mediation, and value alignment represent practical dimensions through which these lenses can be examined.

Adab addresses concerns regarding the learner's relationship with knowledge, teachers, sources, and the learning process. It therefore provides a basis for examining whether GenAI is used to support learning and critical engagement rather than to replace diligent study or teacher guidance. *Amanah* addresses responsibility in the production, use, and dissemination of knowledge. In GenAI-supported IRE, this includes disclosure of AI assistance, verification of generated information, avoidance of plagiarism, protection of learner data, and responsibility for the final use of AI-generated content. These dimensions are consistent with the broader conception of Islamic education as involving not only the acquisition of knowledge but also appropriate intellectual, moral, and social conduct [19, 39].

Epistemic authority addresses a further concern repeatedly identified in the review: how religious knowledge is validated and who is legitimately positioned to interpret it. GenAI may provide explanations, examples, or learning assistance, but it should not be treated as an independent authority on Islamic doctrine, jurisprudence, Qur'anic interpretation, or theological questions. This concern gives rise to the operational dimension of source authenticity, whereby AI-generated religious claims require verification against recognised sources and, where appropriate, mediation by qualified teachers or scholars. Teacher mediation consequently represents the practical mechanism through which AI-generated information can be contextualised, evaluated, and corrected.

These dimensions collectively distinguish the proposed framework from general AI-ethics approaches. General AI governance addresses concerns such as privacy, transparency, fairness, accountability, and academic integrity, whereas the proposed Islamic perspective additionally examines how GenAI-mediated knowledge is approached, who is responsible for it, how religious knowledge is authenticated, and how technology remains aligned with the moral and educational purposes of IRE. The framework should therefore be understood as a provisional, literature-derived analytical framework, rather than a validated model. Its concepts and practical dimensions require further empirical investigation to determine how they can be operationalised and evaluated in actual IRE settings.

5.4 Implications for practice and policy

These findings have several implications for teachers, institutions, and policymakers. For teachers, GenAI can support lesson planning, the generation of examples and formative tasks, and differentiated learning. However, teachers should verify the accuracy of AI-generated materials, provide appropriate religious contextualisation, and educate students about the responsible use of AI. In IRE settings, students should understand that GenAI is an assistive tool rather than a source of religious authority. They should also be encouraged to critically evaluate AI-generated responses, verify relevant information against recognised sources, and disclose their use of GenAI in accordance with institutional requirements.

For institutions, the findings highlight the need for context-specific GenAI policies. Such policies should address academic integrity, disclosure of AI use, appropriate and inappropriate applications, privacy, assessment design, teacher responsibilities, and procedures for verifying AI-generated content. Although general AI policies may provide a useful foundation for IRE, they may not adequately address issues related to religious knowledge, source authenticity, teacher mediation, and Islamic educational values. Institutions providing Islamic education should therefore develop policies that integrate AI literacy with Islamic educational ethics. Teacher training is essential because responsible GenAI integration requires educators to understand its capabilities, limitations, risks, and pedagogical applications [3, 28, 37].

Policymakers and curriculum developers should also establish discipline-specific guidelines for the use of GenAI in Islamic education. Such guidelines should neither unconditionally accept nor reject the technology, but should promote supervised, transparent, and pedagogically appropriate use. Curriculum frameworks should incorporate AI literacy that addresses hallucination, bias, authorship, privacy, plagiarism, source verification, and responsible inquiry. In IRE, these dimensions should be connected explicitly to Islamic values such as *adab* and *amanah*, so that responsible AI use is understood as both an academic and moral responsibility. Policymakers should also address existing gaps in infrastructure and preparedness, particularly uneven access to digital resources and disparities in teacher training [3, 26, 28].

5.5 Evidence boundaries and future research

Despite the insights provided by this review, the evidence base remains limited and should be considered when interpreting the findings and formulating practical implications. The most directly relevant evidence comes from studies of Islamic education, IRE, Islamic teachers, Muslim learners, and Qur'anic education; however, these studies are predominantly exploratory, conceptual, or perception-based. Studies from broader religious education and contextual GenAI ethics provide useful insights into theological, academic-integrity, reliability, and governance issues, but they cannot be treated as direct evidence of GenAI effects in IRE. Accordingly, claims concerning doctrinal distortion, reduced spirituality, or displacement of religious authority should be interpreted as reported concerns or theoretical risks rather than established classroom outcomes. Practical recommendations should similarly be regarded as provisional implications requiring empirical validation.

Further research should therefore move beyond general discussions of the opportunities and risks of GenAI in IRE. Classroom-based empirical studies are needed to

examine how teachers and students actually use these technologies and how their use relates to learning, religious understanding, academic integrity, and teacher–student relationships. Intervention and comparative studies could examine the effects of different forms of teacher guidance and GenAI use across madrasahs, Islamic schools, universities, Qur’anic learning centres, and teacher education programmes. Longitudinal research could further investigate potential changes in religious knowledge, critical thinking, academic integrity, teacher–student relationships, and moral formation over time.

The review also provides a preliminary operationalisation of the proposed Islamic ethical framework. *Adab* may be examined through practices such as critical and respectful engagement with AI-generated religious information, continued engagement with teachers and recognised sources, and avoidance of treating AI output as unquestionable knowledge. *Amanah* may be operationalised through disclosure of AI use, verification of generated content, responsible attribution, protection of learner data, and acceptance of responsibility for AI-assisted work. Epistemic authority may be examined through the extent to which learners and educators verify religious claims against recognised sources and maintain the role of qualified teachers or scholars in interpreting doctrinal, legal, ethical, and theological matters. These dimensions provide starting points for future classroom research, assessment design, institutional guidelines, and GenAI literacy programmes, but require empirical validation before they can be treated as established implementation criteria.

6 Conclusion

This systematic literature review examined the ethical integration of generative artificial intelligence (GenAI) within Islamic Religious Education (IRE). The review found that the available evidence remains limited and is predominantly conceptual, exploratory, or perception-based rather than grounded in classroom implementation. GenAI is generally discussed as an assistive pedagogical technology rather than a substitute for teachers, scholars, or recognised religious sources. The findings therefore support cautious and human-mediated integration rather than claims of established pedagogical effectiveness.

The review further demonstrates that ethical integration in IRE involves both general AI-ethical concerns and distinctive Islamic educational considerations. Issues such as privacy, transparency, bias, plagiarism, and academic integrity remain relevant, but their implications become particularly significant when GenAI is used to produce or mediate religious knowledge. Concerns regarding source authenticity, religious accuracy, teacher mediation, value alignment, *adab*, *amanah*, and epistemic authority therefore extend the discussion beyond conventional AI governance.

The principal theoretical contribution of the review is a preliminary, literature-derived Islamic ethical framework for evaluating GenAI integration in IRE. The framework proposes *adab*, *amanah*, and epistemic authority as core ethical lenses, operationalised through attention to source authenticity, teacher mediation, responsible AI use, and alignment with the purposes of Islamic education. In this framework, *adab* concerns appropriate conduct towards knowledge and learning; *amanah* concerns responsibility for AI-assisted knowledge, authorship, verification, and data; and epistemic authority concerns the legitimate validation and interpretation of religious knowledge. These

dimensions complement rather than replace broader AI-ethics principles by addressing the specific epistemic, moral, and educational purposes of IRE.

This framework should be regarded as provisional rather than a validated model, given the limited evidence base. Future research should therefore empirically operationalise and evaluate these principles through classroom-based, intervention, comparative, and longitudinal studies. Such research can establish whether and how an Islamic ethical approach can support responsible GenAI use while preserving religious knowledge authenticity, teacher mediation, academic integrity, and the moral purposes of IRE.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s44163-026-02251-7>.

Supplementary Material 1

Author contributions

Shahazwan Mat Yusoff conceptualised the study, developed the overall research direction, designed the review framework, and supervised the preparation of the manuscript. Anwar Farhan Mohamad Marzaini contributed to the literature search, screening process, data extraction, thematic organisation, and drafting of the methodology and findings sections. Siti Nadya Zynuddin contributed to the refinement of the research questions, literature synthesis, and critical review of the ethical and pedagogical dimensions of generative AI in Islamic Religious Education. Ahmad Muhaimin Mat Jusoh assisted in data interpretation, thematic analysis, and the development of the discussion section. Mohd Helme Basal contributed to manuscript organisation, academic editing, and refinement of the review structure. Nur Riza Alias supported the literature verification process, reference checking, and language editing. Norhashimi Saad contributed to the critical revision of the manuscript, validation of the review process, and final approval of the manuscript. All authors reviewed, edited, and approved the final version of the manuscript before submission.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Competing interests

The authors declare no competing interests.

Ethical Approval

Not applicable.

Consent for publication

Not applicable. The study does not include any individual-level data or identifiable images requiring consent for publication.

Consent to Participate

Not applicable. This study did not involve any human participants or the collection of primary data requiring informed consent.

Conflict of interest

No conflict of interest is declared by the authors.

Clinical Trial Number

Not applicable.

AI use disclosure statement

During the preparation of this manuscript, generative AI tools were used only to assist with language editing, clarity, and manuscript refinement. The authors critically reviewed and revised all AI-assisted content and remain fully responsible for the accuracy, integrity, analysis, interpretation, and final presentation of the work.

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